

Duy Le

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SUMMARY

Computational research engineer with a PhD focus in physics-informed machine learning and large-scale simulation. Experienced in translating complex scientific problems into scalable AI/ML systems, including end-to-end pipelines from data generation to distributed training and evaluation on HPC GPU clusters. Strong expertise in system design, performance optimisation, and scientific modelling.

RESEARCH INTERESTS

Machine learning for physical systems · Scientific computing · AI for engineering · Computational modelling · Surrogate modelling · Physics-informed ML

TECHNICAL SKILLS

Machine Learning: Deep learning, spatio-temporal models, physics-informed ML
Frameworks: PyTorch (efficient), TensorFlow (familiar)
HPC & Distributed Systems: Slurm, multi-node GPU training, batch scheduling
MLOps: Experiment tracking, reproducible pipelines, evaluation workflows
Programming: Python, NumPy, SciPy
Systems: Linux, HPC environments, data pipelines
Performance: Training optimisation, memory and compute efficiency

RESEARCH & SYSTEMS EXPERIENCE

PhD Researcher – Computational Modelling & Physical AI <i>Federation University / CSIRO Data61</i>	2022 – 2026 <i>Australia</i>
– Designed neural surrogate models for complex physical systems, replacing high-cost numerical simulators while preserving stochastic and physically consistent dynamics. – Built and executed distributed training workflows on HPC GPU clusters using Slurm, enabling scalable learning on large simulation datasets. – Developed end-to-end ML pipelines covering simulation, data processing, model training, and evaluation on HPC systems. – Optimised training and data pipelines for compute efficiency, memory usage, and I/O throughput at scale. – Enforced physical consistency via energy balance, momentum conservation, and system-level diagnostics to ensure stable long-horizon rollouts. – Collaborated with CSIRO researchers on model design, optimisation strategies, and domain-specific evaluation metrics. – Led full lifecycle development from problem formulation and data generation to model scaling, failure analysis, and iterative refinement. – Published first-author papers in peer-reviewed journals and international ML/engineering conferences.	

SELECTED PROJECTS

Neural Surrogate for 3D Granular Flow
Developed a deep learning model using continuous convolution and boundary-motion features to generalise across industrial systems (rotating drum, hopper, blade mixer). Enabled stable long-horizon temporal rollout via multi-stage training.

HPC-Automated Simulation Pipeline
Designed a scalable HPC pipeline integrating physics simulation and ML training, including automated job scheduling, multi-node execution, and validation using Slurm.

Large-Scale Physical Metrics Toolkit
Implemented analysis tools for granular flow statistics, including mixing entropy, alignment metrics, angular velocity fields, kinetic energy, and collision metrics for large-scale evaluation.

PUBLICATIONS

Duy Le, Gary W. Delaney, Linh Nguyen, Truong Phung, David Howard, Gayan Kahandawa, Manzur Murshed, “A Neural Network Surrogate for Modelling Granular Flow Dynamics in Industrial Applications with Dynamic Boundary Conditions,” *Powder Technology*, 2026.

Duy Le, Linh Nguyen, Truong Phung, David Howard, Gayan Kahandawa, Manzur Murshed, Gary W. Delaney, “Machine Learning Accelerated Prediction of 3D Granular Flows in Hoppers,” *The 33rd International Conference on Artificial Neural Networks (ICANN)*, 2024.

Duy Le and Linh Nguyen, “Simple linear iterative clustering based low-cost pseudo-LiDAR for 3D object detection in autonomous driving,” *Multimedia Tools and Applications*, 2023.

Duy Le and Linh Nguyen, “An Efficient Force-Feedback Hand Exoskeleton for Haptic Applications,” *International Journal of Intelligent Robotics and Applications (IJIRA)*, 2021.

Duy Le, Ying Men, Yunkang Luo, Yixuan Zhou and Linh Nguyen, “An Efficient Multi-Vehicle Routing Strategy for Goods Delivery Services,” *IEEE International Conference on Advanced Robotics and its Social Impacts (ARSO)*, 2021.

Duy Le and Linh Nguyen, “A Design of Haptic Hand Exoskeleton for Virtual Reality Applications,” *2021 Innovations in Intelligent Systems and Applications Conference (ASYU)*, 2021.

EDUCATION

Federation University <i>PhD in Computational Modelling & Deep Learning</i> – Thesis: Accelerated Surrogate Modelling of Granular Materials using Artificial Neural Networks – Scholarships: CSIRO Data61 PhD Scholarship, Federation University Tuition Fee Scholarship	Australia <i>2026</i>
Australian National University (ANU) <i>Master of Engineering (Mechatronics)</i> – GPA: 6.2/7.0 (High Distinction)	Australia <i>2021</i>
Ho Chi Minh City University of Technology (HCMUT) <i>Bachelor of Engineering (PFIEV Honours, Mechatronics)</i> – GPA: 3.5/4.0 (Top 3 in PFIEV program)	Vietnam <i>2018</i>

ADDITIONAL INFORMATION

Full working rights in Australia
Experience collaborating across academia and industry
Strong scientific communication (papers, reports, presentations)
Referees available upon request